MILITARY SPECIFICATION

FUZE TIMER, MINE (INTEGRATED CIRCUIT)

1. SCOPE

1.1 This specification covers the inspection and test of one type of CMOS Integrated Circuit (IC) used as the fuze timer in the M56 Anti-tank mine electronics module.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification For.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-883 - Test Methods and Procedures for Micro Electronics
MIL-STD-1168- Lot Numbering of Ammunition.
MIL-STD-1169- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables For Inspection by Attributes

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DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9255865 - Fuze Timer
9278639 - M56 Fuze Timer Static Test Load Card.
9278640 - M56 Fuze Timer Burn - In Load Socket.
9278641 - M56 Fuze Timer Dynamic Test Set.

(Copies of specifications, standards, drawings and publications listed above required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable specifications and drawings.

3.2 Part. - The part shall comply with all requirements specified on drawing (dwg.) 9255865 and with all requirements specified in applicable specifications.

3.3 Pre-Cap Visual, - The IC shall comply with the Pre-cap requirements of MIL-STD-883. Method 2010.1, Test Condition B.

3.4 Bond Strength. - The IC shall comply with the bond strength requirements of MIL-STD-883, Method 2011, Test Condition D.

3.5 Lead, Fatigue. - The IC shall comply with the lead fatigue requirements of MIL-STD-883, Method 2004, Test Condition B2

3.6 Conditioning Screen. - The IC shall be conditioned through the following screening procedures in accordance with Table No. 1 and MIL-STD-883.

TABLE No. 1 CONDITIONING SCREEN REQUIREMENTS

TEST	METHOD	TEST CONDITION
Stabilization Bake	1008	C (48 hours at + 150°C)
Thermal Shock	1011	C (15 cycles)

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TABLE No. 1 CONTINUED

TEST	METHOD	TEST CONDITION
Temperature Cycling	1010	C (10 cycles)
Mechanical Shock	2002	B (5 Pulses, Y1 direction)
Centrifuge	2001	E (Y 1 direction only)
Fine Leak	1014	A
Gross Leak	1014	C

3.7 Pre-Burn - In Static Electrical Parameters. - The IC shall comply with the requirements of Table NO. 2 when the device is tested at a temperature of twenty five (25) plus or minus five degrees centigrade (25 ±~ 5°C) using the test equipment of 4.4.4.

TABLE NO. 2
STATIC AMBIENT ELECTRICAL PARAMETERS

PIN NO.	PARAMETER (FUNCTION)	FUNCTION No. 1	FUNCTION No. 2	FUNCTION No. 3
1	(VSS)	G	G	G
2	(IH)	G	G	D
3	(VDD)	7.0V	7.0V	7.0V
4	(C)	F		
5	(R)	E		
6	(RC)	D	C	C
7	(OR)	C	B	B
8	(AD)	B	G	G
9	(AR)	A	A	A
10	(AE)	G	G	G
11	(DE)	G	G	G
12	(DR)	G	G	G
13	(BD)			
14	(RZ)			
15	(D)		D	E
16	(VC)			
LIMITS				
MINIMUM		4.85	0.4	0.4
MAXIMUM		2.15	0.20	0.20
UNIT		VOLTS	VOLTS	VOLTS
NOTES		1	2	3

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TABLE NO. 2 CONTINUED

PIN NO. PARAMETER	FUNCTION No. 4	DISSIPATION CURRENT # 1	DISSIPATION CURRENT # 2	DISSIPATION CURRENT # 3	DISSIPATIO CURRENT #
1	G	G	G	G	G
2	G	7.0V	7.0V	G	8.0V
3	7.0V	TEST 7.0V	TEST 7.0V	TEST 7.0V	TEST 8.0V
4					
5					
6	C	G	G	C	G
7	B	G	G	B	G
8	G	G	G	G	G
9	A			A	
10	G	G	1.0V	G	1.0V
11	G	G	G	G	G
12	G				
13					
14					
15	D				
16					
MAXIMUM	0.4	260	190	190	850
MINIMUM	0.20				
UNIT	VOLTS	MICRO-AMPERES	MICRO-AMPERES	MICRO-AMPERES	MICRO-AMPERES
NOTES	4	-	6	5	6

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TABLE NO. 2 CONTINUED

PIN/PARAMETER	ARMING E-Cell "ON"-1	ARMING E-Cell "ON"-2	DELAY E-Cell "ON"-1	DELAY E-Cell "ON"-2	AEC-DEC "OFF" -1
1	G	G	G	G	G
2	G	G	G	G	G
3	6.75V	6.75V	6.75V	6.75V	6.75V
4	6.75V	6.75V	6.75V	6.75V	6.75V
5	G	G	G	G	G
6	G	G	G	G	G
7	G	G	G	G	G
8	G	G	G	G	G
9					
10	TEST 0.15V	TEST 0.15V	G	G	
11	G	G	TEST 0.15V	TEST 0.15V	TEST 0.7V
12				0.35V	
13					
14	6.75V	6.75V	6.75V	6.75V	6.75V
15					
16					
LIMITS					
MINIMUM	-120		-24		-10
MAXIMUM		-60		-20	
UNITS	MICRO-AMPERES	MICRO-AMPERES	MICRO-AMPERES	MICRO-AMPERES	MICRO-AMPERES

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TABLE NO. 2 CONTINUED

PIN/PARAMETER	AEC-DEC "OFF"-2	Pgm	Ngm	ICH-1	ICH-2
1	G	G	G	G	G
2	G	G	G	G	G
3	6.75V	6.75	6.75V	6.75V	6.75V
4	6.75V	4.75V TEST	2.0V TEST	6.75V	6.75V
5	G	G	6.75V	G	G
6	G	6.75V	G	G	G
7	G	G	6.75V	G	G
8	G	G	G	G	G
9					
10	0.7V TEST	G	G	1.0V	1.0V
11		G	G	1.0V	1.0V
12					
13					
14	6.75V	6.75V	6.75V	6.75V	6.75V
15					TEST
16				G TEST	G
LIMITS					
MINIMUM	-10		200		
MAXIMUM		-200		-25	0.1
UNITS	MICRO-AMPERES	MICRO-AMPERES	MICRO-AMPERES	MICRO-AMPERES	VOLTS

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TABLE NO. 2 CONTINUED

PIN/PARAMETER	ARMING E-Cell FUNCTION	DELAY E-Cell FUNCTION	LVD NO-FIRE	LVD FIRE	SCR V-1
1	G	G	G	G	G
2	G	G	G	G	G
3	6.75V	6.75V	6.00	5.40	6.75V
4	6.75V	6.75V			
5	G	G			
6	G	G	G	G	G
7	G	G	G	G	G
8	G	G			
9					
10	1.0V	1.0V	1.0V	1.0V	1.0V
11	G	1.0V	G	G	1.0V
12					
13					
14	6.75V	6.75V			
15		TEST	TEST	TEST	G
16	TEST				50 ma TEST
LIMITS					
MINIMUM	5.50	1.50		0.7	
MAXIMUM			0.6		1.6
VOLTS	VOLTS	VOLTS	VOLTS	VOLTS	VOLTS
NOTES	6	7	6	6	7,9

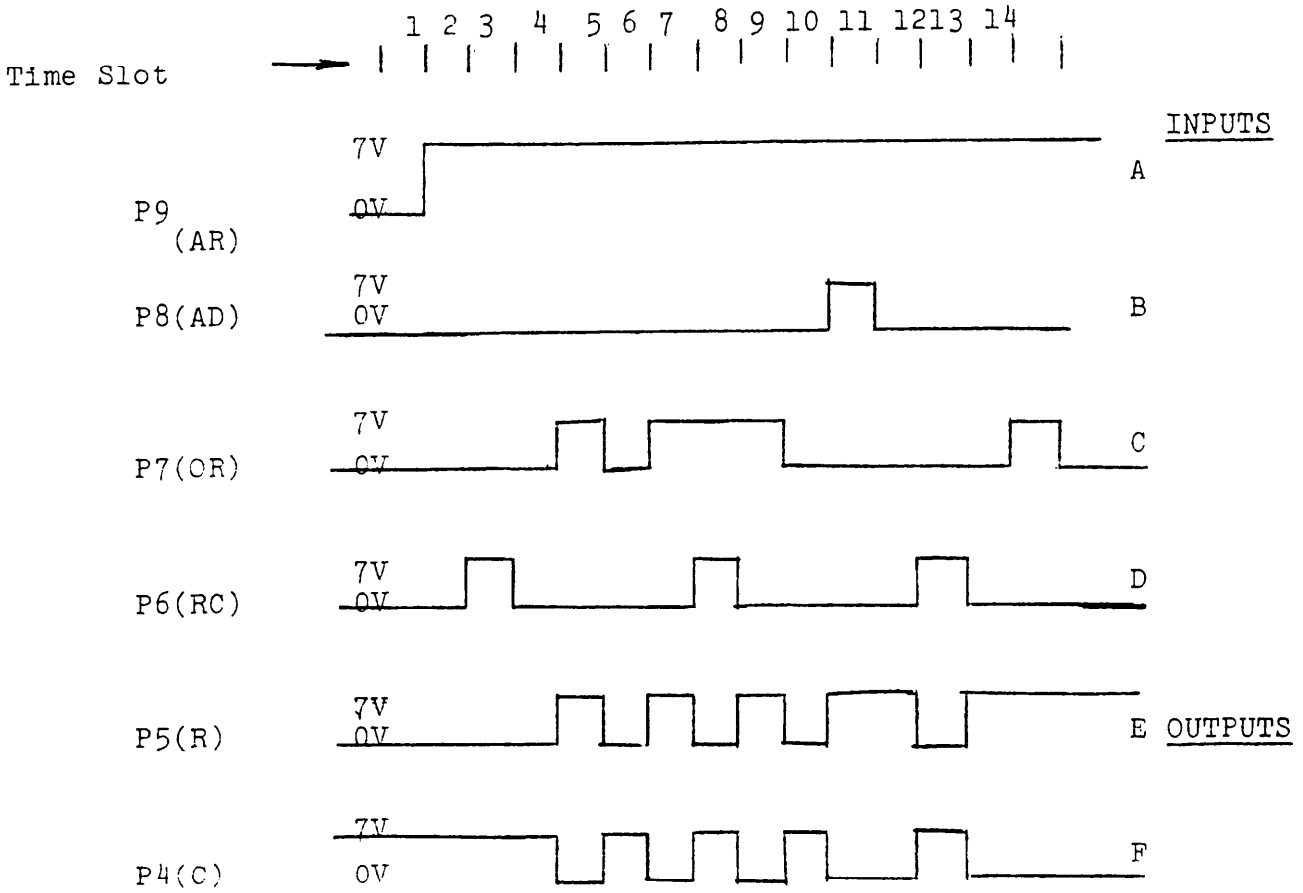
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TABLE NO. 2 CONTINUED

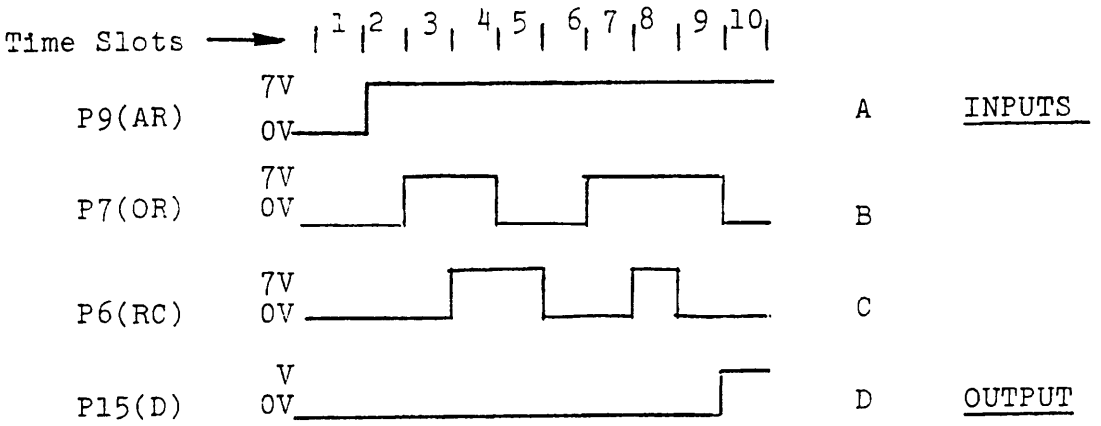
PIN # / PARAMETER	SCR V-2	DELAY E-CELL FUNCTION
1	G	G
2	G	G
3	6.75V	6.75V
4		6.75V
5		G
6	G	G
7	G	G
8		G
9		
10	1.0V	1.0V
11	1.0V	1.0V
12		
13		
14		6.75V
15	G	TEST
16	100 ma TEST	
LIMITS		
MINIMUM		1.50
MAXIMUM	0.25	
UNITS	VOLTS	VOLTS
NOTES	7,9	7

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NOTE 1: Function 1 - Functional Test consisting of 16 time slots. Inputs are driven to levels Indicated. Outputs must correspond to levels indicated.



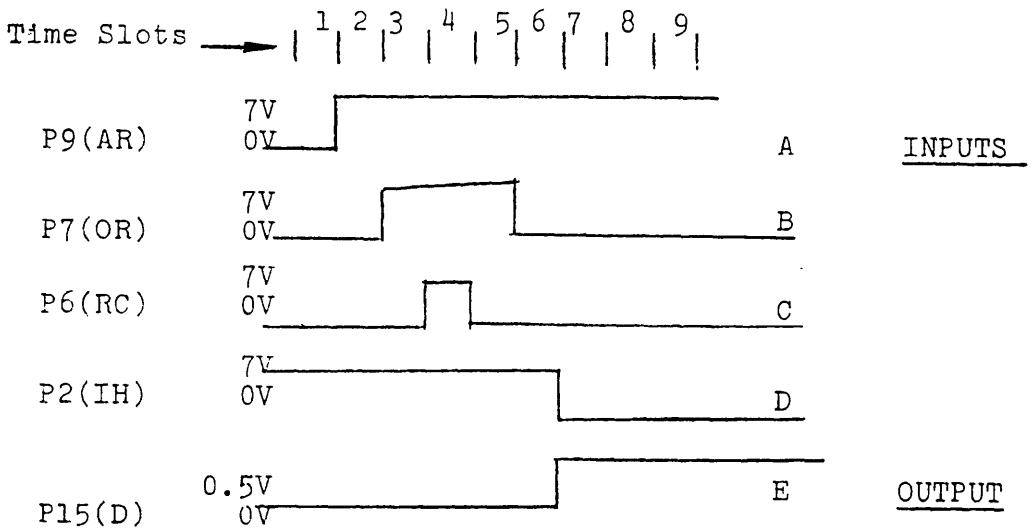
NOTE 2 :



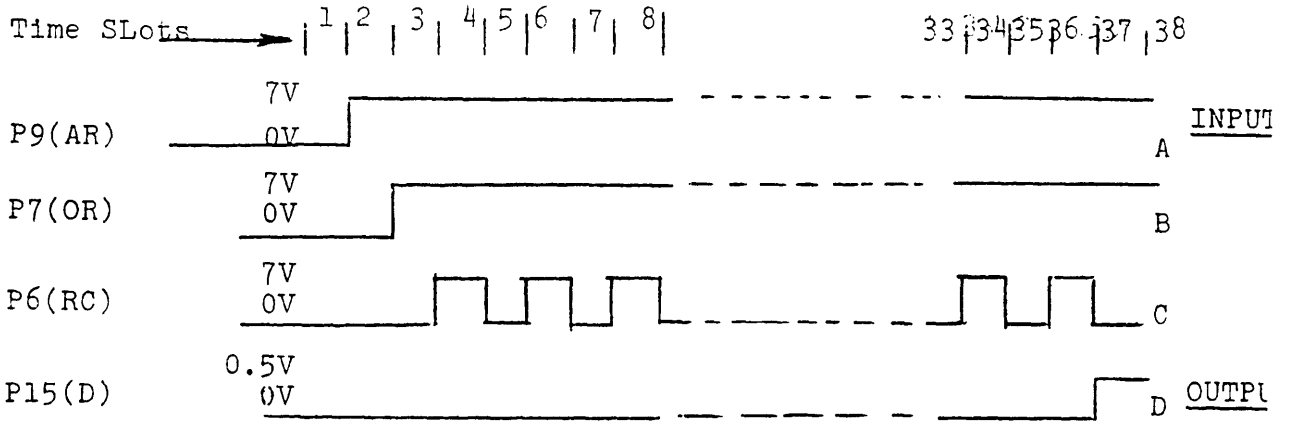
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Function 2 - Functional Test consisting of 10 time slots. Inputs are driven to levels indicated. Outputs must correspond to levels indicated.

NOTE 3: Function 3 - Functional Test consisting of 7 time slots. Inputs are driven to levels indicated. Outputs must correspond to levels indicated.



NOTE 4: Function 4 - Functional Test consisting of 37 time slots. Inputs are driven to levels indicated. Outputs must correspond to leve is indicated.



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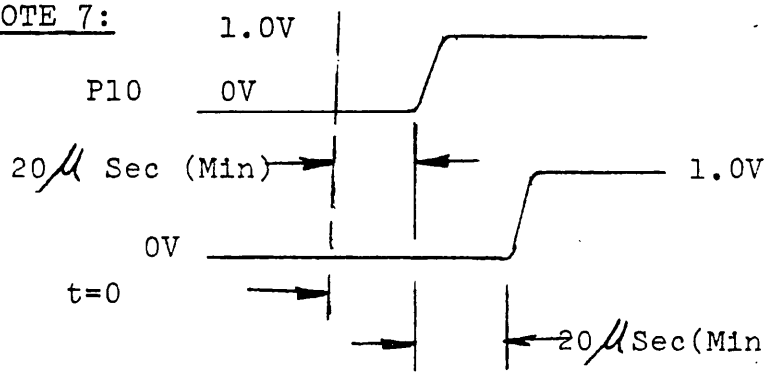
NOTE 5: Test 7 - Disspation 3

Input waveforms same as note 4 except:

a) Output D not required
b) Stop in time slot 33 and make required measurement (all 1's dissipation).

NOTE 6: 1.0 volts on P 10 delayed 20 microseconds (minimum) from other input levels.

NOTE 7:



All other input levels established

NOTE 8: For all tests, power-on sequence is as follows:

1 - All specified input drives must be connected to GND.
2 - VDD (VCC1 or MS) is then applied.
3 - Input high levels then established
4 - Measurement may be made.

NOTE 9: Close Relay and Shunt 5K for these tests only.

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3.8 Burn-In. - The IC shall be burned-in at a temperature of one hundred and twenty five degrees centigrade plus or minus five degrees centigrade (125°C ± 5°C) for a minimum time of one hundred and sixty eight (168) hours using the test equipment of 4.4.5.

3.9 Group A Final Static Ambient Electrical Parameters. - The IC shall meet the requirements of Table No. 2 when it is tested at a temperature of twenty five degrees centigrade plus or minus five degrees centigrade (25°C ± 5°C) using the test equipment of 4.4.6.

3.10 Group A Final High Temperature Static Electrical Parameters. - The IC shall meet the requirements of Table No. 3 when it is tested at a temperature of one hundred and twenty five degrees centigrade plus or minus five degrees centigrade (125°C ± 5°C) using the equipment of 4.4.7.

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TABLE NO. 3

GROUP A FINAL STATIC HIGH TEMPERATURE ELECTRICAL PARAMETERS

PIN #/PARAMETER	FUNCTION NO. 1	FUNCTION NO. 2	FUNCTION NO. 3	FUNCTION NO. 4
1 (VSS)	G	G	G	G
2 (IH)	G	G	D	G
3 (VDD)	7.0V	7.0V	7.0V	7.0V
4 (C)	F			
5 (R)	E			
6 (RC)	D	C	C	C
7 (OR)	C	B	B	B
8 (AD)	B	G	G	G
9 (AR)	A	A	A	A
10 (AE)	G	G	G	G
11 (DE)	G	G	G	G
12 (DR)	G	G	G	G
13 (BD)				
14 (RZ)				
15 (D)		D	E	D
16 (VC)				
LIMITS				
MINIMUM	4.45	0.43	0.43	0.43
MAXIMUM	2.35	0.17	0.17	0.17
UNITS	VOLTS	VOLTS	VOLTS	VOLTS
NOTES	1	2	3	4

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TABLE NO. 3 CONTINUED

PIN #/PARAMETER	DISSIPATION CURRENT # 2	AEC-DEC "OFF"-1	AEC-DEC "OFF"-2	SCR V-1	SCR V-2
1	G	G	G	G	G
2	7.0V	G	G	G	G
3	7.0V TEST	6.75V	6.75V	6.75V	6.75V
4		6.75V	6.75V		
5		G	G		
6	G	G	G	G	G
7	G	G	G	G	G
8	G	G	G		
9					
10	1.0		0.7V TEST	1.0V	1.0V
11	G	0.7V TEST		1.0V	1.0V
12					
13					
14		6.75V	6.75V		
15				G	G
16				50 ma TEST	100 ma TEST
LIMITS					
MINIMUM		- 25	- 25		
MAXIMUM	550			1.6	0.25
UNITS	MICRO-AMPERES	MICRO-AMPERES	MICRO-AMPERES	VOLTS	VOLTS
NOTES	6			7,9	7,9

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3.11 Group A Final Low Temperature Static Electrical Parameters. - The IC shall meet the requirements of Table No. 4 when it is tested at a temperature of minus fifty five degrees centigrade plus or minus five degrees centigrade (-55°C ± 5°C) using the test equipment of 4.4.8.

TABLE NO. 4

GROUP A FINAL STATIC LOW TEMPERATURE ELECTRICAL PARAMETERS

PIN #/PARAMETER	FUNCTION No. 1	FUNCTION No. 2	FUNCTION No. 3	FUNCTION No. 4
1 (VSS)	G	G	G	G
2 (IH)	G	G	D	G
3 (VDD)	7.0V	7.0V	7.0V	7.0V
4 (C)	F			
5 (R)	E			
6 (RC)	D	C	C	C
7 (OR)	C	B	B	B
8 (AD)	B	G	G	G
9 (AR)	A	A	A	A
10 (AE)	G	G	G	G
11 (DE)	G	G	G	G
12 (DR)	G	G	G	G
13 (BD)				
14 (RZ)				
15 (D)		D	E	D
16 (VC)				
LIMITS				
MINIMUM	5.3	0.53	0.53	0.53
MAXIMUM	2.0	0.15	0.15	0.15
UNITS	VOLTS	VOLTS	VOLTS	VOLTS
NOTES	1	2	3	4

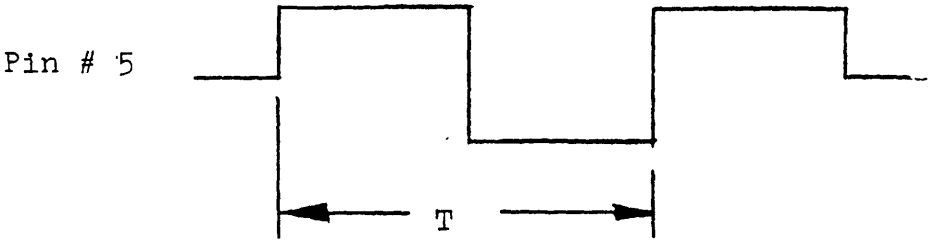
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TABLE NO. 4 CONTINUED

PIN #/PARAMETER	DISSIPATION CURRENT # 2	AEC-DEC "OFF"-1	AEC-DEC "OFF"-2	SCR V-1	SCR V-2
1	G	G	G	G	G
2	7.0V	G	G	G	G
3	7.0V TEST	6.75V	6.75V	6.75V	6.75V
4		6.75V	6.7V		
5		G	G		
6	G	G	G	G	G
7	G	G	G	G	G
8	G	G	G		
9					
10	1.0V		0.7V TEST	1.0V	1.0V
11	G	0.7V TEST		1.0V	1.0V
12					
13					
14		6.75V	6.75V		
15				G	G
16				50 ma TEST	100 ma TEST
LIMITS					
MINIMUM		- 10	- 10		
MAXIMUM	185			1.6	0.25
UNITS	MICRO-AMPERES	MICRO-AMPERES	MICRO-AMPERES	VOLTS	VOLTS
NOTES	6			7,9	7,9

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3.12 Dynamic Parameter. - The IC shall have the following output wave form when tested in accordance with 4.4.9 at a temperature of twenty five plus or minus ten degrees centigrade (25°C ± 10°C).



$16 \leq T \geq 25$ Milliseconds

3.13 Group B Environmental Requirements. - The IC shall comply with the requirements of Table No. 5 and MIL-STD-883.

TABLE NO. 5

GROUP B ENVIRONMENTAL TESTS

TEST	METHOD	TEST CONDITION
Physical Dimensions	2008	A
Marking Permanency	2008	B (Per 3.2.1)
Visual and Mechanical	2008	B (10X)
Bond Strength	2011	D
Solderability	2003	
Lend Fatigue	2004	B2 (Any 5 Leads)
Fine Leak	1014	A
Gross Leak	1014	C

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3.14 Group C Environmental Requirements. - The IC shall comply with the requirements of Table No. 6 and MIL-STD-883

TABLE NO. 6

GROUP C ENVIRONMENTAL REQUIREMENTS

TEST	METHOD	TEST CONDITION
Thermal Shock	1011	C
Temperature Cycling	1010	C
Moisture Resistance	1004	No Voltage Applied
Fine Leak	1014	A
Gross Leak	1014	C
Mechanical Shock	2002	B (0.5 ms)
Vibration Variable Frequency	2007	A
Constant Acceleration	2001	E
Salt Atmosphere	1009	A (without initial conditioning)
High Temperature Storage	1008	C (1000 hours)
Operating Life (Bias Condition)	1005	+ 125°C for 1000 hours
Electrical Tests per 3.9	-	-

3.15 Test Programming. - All final static electrical acceptance testing shall be performed using Government tapes and load cards described by 4.4.6, 4.4.7, and 4.4.8.

3.16 Workmanship. - All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning agent or method *used* shall not be injurious to any part, nor shall the part be contaminated by the cleaning agent. All markings shall be neat and sharply defined.

3.17 Submission of Results of Contractor Conducted Examinations and Tests. - The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. Results shall be submitted within thirty (30) days at the completion of each lot tested (see 6.4).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform *any* of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and servies conform to the prescribed requirements. Reference shall be made to MIL-STD-109 *to* define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is-correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.3 Inspection Provisions

4.3.1 Lot Formation. - The term “inspection lot” as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MTL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity’ criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent, Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by theGovernment. Instead of applying the AQL’s designated herein to the groups of major and minor defects listed, the contractor may elect to apply an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of’ the major and minor defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor’s inspection system. Equipment necessary for the performance of the inspecitons listed shall be in accordance with 4.3.4.

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4.3.2.1 Pre-Cap Visual (dwg. 9255865).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major:	100 percent		
101.	MIL-STD-883, Method 2010.1, Test Condition B.....	Visual	01001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.16)	Visual	01002

4.3.3 TESTING

4.3.3.1 Bond Strength, (see 3.4) Major Defect, Code No. 10001. - Twenty (20) IC's shall be randomly selected from each lot and tested in accordance with 4.4.1. The lot shall be rejected if one (1) or more IC'S are classed defective.

4.3.3.2 Lead Fatigue (see 3.5) Major Defect, Code No. 11002. - Six (6) IC's shall be randomly selected from each hour's production. That hour's production shall be classed defective, if one (1) or more of the devices are classed defective, Testing shall be in accordance with 4.4.2.

4.3.3.3 Conditioning Screen Tests, (see 3.6) Major Defects, The conditioning tests of Table No. 7 shall be performed in the sequence shown in accordance with Table No. 7 and 4.4.3.

TABLE NO. 7 CONDITIONING SCREEN DEFECTS

Code No.	Test
12001	Stabilization Bake
12002	Thermal Shock
12003	Temperature Cycling
12004	Mechanical Shock
12005	Centrifuge
12006	Fine Leak
12007	Gross Leak

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Each IC lot shall be screened through these test 100 percent prior to being submitted to the initial electrical test. Any IC failing to pass all of the above conditioning tests shall be classed defective and removed from the lot. The Government shall monitor these tests during their performance to assure that they comply with all of the specified requirements of 3.6.

4.3.3.4 Pre-Burn-In Static Electrical Parameter Tests, (see 3.7), Major Defects. - Each lot shall be tested 100 percent for the Pre-Burn-In Static electrical parameters of Table No. 8 in the sequence shown and in accordance with 4.4.4, after passing the condition screen tests of 4.3.3.3.

TABLE NO. 8
PRE-BURN-IN STATIC AMBIENT ELECTRICAL PARAMETER TESTS

Code No.	Test
13001	Function No. 1
13002	Function No. 2
13003	Function No. 3
13004	Function No. 4
13005	Dissipation Current No. 1
13006	Dissipation Current No. 2
13007	Dissipation Current No. 3
13008	Dissipation Current No. 4
13009	Arming E-Cell "ON" - 1
13010	Arming E-Cell "ON" - 2
13011	Delay E-Cell "ON' - 1
13012	Delay E-Cell "ON" - 2
13013	AEC-DEC "OFF" - 1
13014	AEC-DEC "OFF" - 2
13015	Pgm
13016	Ngm
13017	ICH-1
13018	ICH-2
13019	Arming E-Cell Function
13020	Delay E-Cell Function
13021	LVD No-Fire
13022	LVD Fire
13023	SCR V-1
13024	SCR V-2
13025	Delay E-Cell Function

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Any IC classed defect shall be removed from the lot. The Government shall monitor the contractor's tests to the extent necessary to assure compliance with the requirements of 3.7.

4.3,3.5 Burn-In, (see 3.8) Major Defect, Code No. 14001. Each lot of IC's following the pre-burn-in electrical testing of 4.3.3.4 shall be tested 100% for the specified requirements of 3.8 in accordance with 4.4.5. The lot of IC's, after completing the burn-in requirements, shall be subjected to final electrical tests. The Government shall monitor the contractor's facilities to the extent necessary to assure compliance with the requirements of 3.8.

4.3.3.6 Group A Final Static Ambient Electrical Parameters (see 3.9) Major Defects. - Each lot of IC's shall be tested 100 percent for the requirements listed in Table No. 9 in sequential order and in accordance with 4.4,6. Any IC classed defective shall be removed from the lot.

TABLE 9

GROUP A FINAL STATIC AMBIENT ELECTRICAL PARAMETERS

Code No.	Defect
15001	Function No. 1
15002	Function No. 2
15003	Function No. 3
15004	Function No. 4
15005	Dissipation Current No. 1
15006	Dissipation Current No. 2
15007	Dissipation Current NO. 3
15008	Dissipation Current No. 4
15009	Arming E-Cell "ON" - 1
15010	Arming E-Cell "ON" - 2
15011	Delay E-Cell "ON" - 1
15012	Delay E-Cell "ON" - 2
15013	AEC-DEC "OFF" - 1
15014	AEC-DEC "OFF" - 2
15015	Pgm
15016	Ngm
15017	ICH-1

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TABLE NO. 9 CONTINUED

Code No.	Defect
15018	ICH-2
15019	Arming E-Cell Function
15020	Delay E-Cell Function
15021	LVD No-Fire
15022	LVD Fire
15023	SCR- V-1
15024	SCR- V-2
15025	Delay E-Cell Function

4.3.3.7 Group A Final Static High Temperature Electrical Parameters (see 3.10) Major Defects. - Each lot of IC'S shall be tested 100 percent after passing the ambient electrical tests for the requirements listed in Table No. 10 in sequential order and in accordance with 4.4.7. Any IC classed defective shall be removed from the lot.

TABLE NO. 10

GROUP A FINAL STATIC HIGH TEMPERATURE ELECTRICAL PARAMETERS

Code No.	Defect
16001	Function No. 1
16002	Function No. 2
16003	Function No. 3
16004	Function No. 4
16005	Dissipation Current No. 2
16007	AEC-DEC "OFF" - 1
16008	AEC-DEC "OFF" - 2
16009	SCR V-1
16010	SCR V-2

4.3.3.8 Group A Final Static Low Temperature Electrical Parameters (see 3.11) Major Defects. - Each lot of IC's shall be tested 100 percent after passing the high temperature electrical tests for the requirements of Table No. 11 in sequential order

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and in accordance with 4.4.8. Any XC classed defective shall be removed from the lot. After testing, those IC's that are not classed defective shall be sent to the dynamic tests of 4.3.3.9.

TABLE NO. 11

GROUP A FINAL STATIC LOW TEMPERATURE ELECTRICAL PARAMETERS

Code No.	Defect
17001	Function No. 1
17002	Function No. 2
17003	Function No. 3
17004	Function No. 4
17005	Dissipation Current No. 2
17006	AEC-DEC "OFF" - 1
17007	AEC-DEC "OFF" - 2
17008	SCR V-1
17009	SCR V-2

4.3,3.9 Dynamic Parameter (see 3.12) Major Defect, Code No. 18001. The lot shall be screened 100 percent following the static tests for the specified requirement in accordance with 4.4.9. Any IC classed defective shall be removed from the lot.

4.3.3.10 Group B Environment Tests, (see 3.13) Major Defects. - The IC's shall be tested for the requirements of Table NO. 12 once every six (6) weeks on a continuous basis while the device is being produced. Testing shall be in accordance with MIL-STD-883 and 4.4,10.

TABLE NO. 12

GROUP B ENVIRONMENTAL DEFECTS

Code No.	Sub-Group	Test	Quantity	Allowable Defects
19001	1	Physical Dimension	25	1
19002	2	Marking Permanency	4	0

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TABLE NO. 12 (CONTINUED)

GROUP B ENVIRONMENTAL DEFECTS

Code No.	Sub-Group	Test	Quantity	Allowable Defects
19003		Visual and Mechanical	25	1
19004		Bond Strength		
19005	3	Solderability	25	1
19006	4	Lead Fatigue	25	1
19007		Fine Leak		
19008		Gross Leak		

Further lots shall not be shipped if the devices of any sub-group fail to comply with all of the specified requirements.

4.3.3.11 Group C Environment Tests, Major Defects. - The IC's shall be tested for the requirements of Table No. 13 on a continuing schedule while the device is being produced. Testing shall be in accordance with MIL-STD-883 and 4.4.11.

TABLE NO. 13

GROUP C ENVIRONMENTAL DEFECTS

Code No.	Sub-Group	Test	Quantity	Allowable Defects
20001	1	Thermal Shock	25	1
20002		Temperature cycling		
20003		Moisture Resistance		
20004		Fine Leak (After moisture resistance)		
20005		Gross Leak (After moisture resistance)		

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TABLE NO. 13 CONTINUED

Code No.	Sub-Group	Test	Quantity	Allowable Defects
20006		Electrical Test per 3.9		
20007	2	Mechanical Shock	25	1
20008		Vibration, Variable Frequency		
20009		Constant Acceleration		
20010		Fine Leak		
20011		Gross Leak		
20012		Electrical Test Per 3.9		
20013	3	Salt Atmosphere	25	1
20014	4	High Temperature Storage	25	1
20015		Electrical Test per 3.9		
20016	5	Operating Life	77	0
20017		Electrical Tests per 3.9		

Further lots shall not be shipped if the devices fail to

4.3.4 Inspection Equipment. - Inspection equipment required to perform the inspectionprescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection design and approvcal.

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4.4.1 Bond Strength. - The IC shall be tested for the requirements of Method 2011.1, Test Condition D of MIL-STD-883 using Government approved test equipment. Any IC failing to comply with all of the requirements of 3.4 shall be classed defective. Testing is destructive and devices tested shall not be returned to the lot.

4.4.2 Lead Fatigue. - The IC shall be tested for the requirements of Method 2004, Test Condition B2 of MIL-STD-883 using Government approved test equipment. Any IC failing to comply with all of the requirements of 3.5 shall be classed defective. Testing is destructive and devices classed defective shall not be returned to the lot,

4.4.3 Conditioning Screen Tests. - The IC shall be subjected to the required tests using only Government approved test equipment. Any IC failing to comply with all of the requirements of 3.6 shall be classed defective. Testing is non-destructive and only IC's that pass all of the specified tests shall be returned to the lot.

4.4.4 Pre-Burn-In Static Electrical Parameter Tests. - The IC shall be tested for the specified requirements of 3.7 using a Teredyne IC Test Set No. J283 with H260 test check with 36 channel capacity, and the load card of drawing 9278639. The test conditions and limits shall be placed on a Tri-Data 10 foot tape No. 1010 or equivalent tape compatible with the test equipment. Any IC failing to comply with all of the specified requirements of 3.7 shall be classed defective. Testing is non-destructive and only accepted IC's shall be returned to the lot.

4.4.5 Burn-In Tests. - The IC shall be tested for the specified requirements of 3.8 using approved tests racks and sockets in accordance with drawing 9278840. Testing is non-destructive and IC's shall be sent to final electrical testing of 4.3.3.6.

4.4.6 Group A Final Static Ambient Electrical Parameters. - the IC shall be tested for the specified requirements of 3.9 using a Teredyne IC Test Set No. J283 with H360 test check with 36 channel 'capacity and the load card of drawing 9278639. The test conditions and limits shall be placed on a Tri-Data 10 foot tape No. 1010 or equivalent tape compatible with the test equipment.

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Any IC failing to comply with all of the specified requirements of 3.9 shall be classed defective. Testing is non-destructive and only accepted IC's shall be returned to the lot.

4.4.7 Group A Final Static High Temperature Electrical Parameters. - The IC shall be tested for the specified requirements of 3.10 using a Teredyne IC Test Set NO.- J283 with H360 test check and 36 channel capacity and the load card of drawing 9278639. The test conditions and limits shall be placed on a Tri-Data 10 foot tape No. 1010 or equivalent tape compatible with the test equipment. Any IC failing to comply with all of the specified requirements of 3.10 shall be classed defective. Testing is non-destructive and only accepted IC's shall be returned to the lot.

4.4.8 Group A Final Static Low Temperature Electrical Parameters. - The IC shall be tested for the specified requirements of 3.11 using a Teredyne IC Test Set No. J283 with H360 test check and 36 channel capacity and the load card of drawing 9278639. The test conditions and limits snail be placed on a Tri-Data 10 foot tape No. 1010 or equivalent tape compatible with the test equipment. Any IC failing to comply with all of the specified requirements of 3.11 shall be classed defective. Testing is non-destructive and only accepted IC's shall be returned to the lot.

4.4,9 Dynamic Parameter Tests. - The IC shall be tested using the Dynamic Test Set, dwg. 9278641 for the specified requirements of 3.12. Any IC failing to comply with all of the requirements of 3.12 shall be classed defective and removed from the lot. Testing is non-destructive and only accepted IC's shall be returned to the lot.

4.4.10 Group B Environmental Tests. - The IC shall be tested using Government approved test equpment and procedures. Any IC failing to comply with all of the specified requirements of 3.13 shall be classed defective. Tested IC's shall not be returned to the lot.

4.4.11 Group C Environmental Tests. - The XC shall be tested using Government approved test equipment and procedures. Any IC failing to comply with all of the specified requirements of 3.14 shall-be classed defective. Tested IC's shall not be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 IC Conductive Packaging. - Each IC shall be mounted into Emmerson and Cummings, Inc. "ECCOSORB" conductive material or equivalent prior to packaging.

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5.1.2 Level C Packaging. - The IC after being mounted in the conductive material of 5.1.1, shall be packed for shipment in accordance with MIL-STD-1169.

5.2 Marking. - Marking of the carton shall comply with MIL-STD-1169.

5.3 Shipping. - When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for eac lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code number assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equip ment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the “Method of Inspection” headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs. -SIE designs-may consist of any of the following:

a. Detailed drawings which completely depict all infor mation necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

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c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall heet. l When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. -Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations-and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered hereby (see 6.3.1c). Designs shall be of the category and form per MIL-D-1000) specified in the Contract Data Requirements Lists DD Form 1423). The contractor is referred to MIL-HDBK-204, Inspection Equipment Design" for guidance. The specification umber and the applicable five digit defect code number (or ther specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together ith the component or assembly drawing number and revision letter o which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs hall be approved by the Government prior-to fabricating or pro- uring the equipment. Designs shall be submitted for approval to icatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 n accordance with the stipulations, time frame and distribution pecified in the Contract Data Requirements List (DD Form 1423) r in the contract. The contractor must include in his sub- ission of' designs a listing of any and all Government SIE esigns which will be utilized. Partial submission of inspec- ion designs is permissible and encouraged. However, the ompletion date for design review will be based on the date f the final submission of designs. Picatinny Arsenal design eview will normally be accomplished within one month after eceipt.

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6.4 Submission results of contractor conducted examinations and tests (see 3.17). - The contractor shall send his results to:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-M
Dover, New Jersey 07801

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1345-A-031

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

I. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

. REMARKS

UBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

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